# Dasari Naga Raju

*Project Research Assistant at Medical College of Wisconsin, USA*

Andhra Pradesh, India | raajuuu1998@gmail.com | **Project TARAA** | | | |

## Research Interests

My research interests center on biologically grounded representation learning for molecular biomarker and phenotype prediction from H&E whole-slide images, using pathology foundation models and weakly supervised learning. I am particularly interested in improving generalization under cross-cancer, cross-site, and combined distribution shifts, while distinguishing transferable biological patterns from cancer-specific signals.

## Education

**International Institute of Information Technology Bangalore** *Sept 2021 – July 2024*
M.S. by Research in Information Technology CGPA: 3.16 / 4.0
*Research Area: Computational Pathology | 59/75 credits dedicated to supervised thesis research*

**Jawaharlal Nehru Technological University Gurajada, Vizianagaram** *July 2016 – Sept 2020*
B.Tech. in Electronics and Communication Engineering CGPA: 7.6 / 10

## M.S. Thesis

**DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology** *Published at MICCAI 2026 Workshop on BrainWorks* [Paper]

- Demonstrated that explicitly encoding nuclear spatial patterns provides discriminative information beyond tissue appearance and general-purpose pathology foundation models in data-limited settings; DeepNucleiNet achieved a mean **F1-score of 0.942**, compared with **0.884** for UNI2-h and **0.730** for Virchow2.
- Evaluated cross-tumour generalization without retraining, achieving an **F1-score of 0.943** on TCGA-GBM, **0.843** on DeepHisto, and **0.635** on TiGER. The model produced no necrosis predictions across **85,632** TCGA-LGG negative-control tiles.

## Independent Computational Pathology Research

**Tissue-Aware Representation Adaptation Across Cancers (Project TARAA)** *Jan 2025 – Present*

Through Project TARAA, I independently develop and evaluate methods for molecular biomarker prediction from H&E images across cancer types. My current research investigates hyperbolic geometry and curvature-aware representation learning to address cross-organ and cross-site distribution shifts, separating transferable biological patterns from organ-specific signals. My recent work, TIRA (Tumor Immune Representation Adaptation), aims to improve cross-cancer and cross-organ generalization for MSI and TMB biomarker prediction **[Paper]**.

## Publications

1. **Dasari Naga Raju**, Tripti Bameta, "State-Aware Interaction MIL for Rare Joint Molecular Phenotype Prediction in Colorectal Cancer and Lung Adenocarcinoma." *Accepted at **NeurIPS 2026 Workshop** on AISC: Cancer Pathology Foundation Models.* **[Paper] [Code]**
2. **Dasari Naga Raju**, "Conserved Immune Topology Improves Pathology Foundation Model Generalization for Cross-Cancer MSI-H Prediction." *Published at **ECCV 2026 Workshop** on Medical Foundation Models and Benchmarks (MedFMB).* **[Paper] [Code]**
3. **Dasari Naga Raju**, "What Do Pathology Foundation Models Learn Beyond Gleason Grade in Prostate Cancer?" *Published at **MICCAI 2026 Workshop** on Computational Pathology and Multimodal Data (COMPAYL).* **[Paper] [Code]**
4. **Dasari Naga Raju**, T.K. Srikanth, Shilpa Rao, Ramesh Kestur, Anita Mahadevan, "DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology." *Published at **MICCAI 2026 Workshop** on BrainWorks.* **[Paper] [Code]**
5. **Dasari Naga Raju**, V.K. Kancharla, P. Kaveti, "Surgical Foundation Model Ensembles for Predicting Annotator Disagreement in Critical View of Safety Assessments." *Published at **MICCAI 2026 Workshop** on Clinical-driven Robotics and Embodied AI Technology (CREATE).* **[Paper]**
6. V.K. Kancharla, **Dasari Naga Raju**, P. Kaveti, "Self-Supervised Denoising Reconstruction Outperforms Foundation Models for Capsule Endoscopy Abnormality Classification." *Published at **MICCAI 2026 Workshop** on AI for Safe Surgery (SafeSurg).* **[Paper]**
7. Pavan Kaveti, Vamshi Krishna Kancharla, **Dasari Naga Raju**, "Weave Before You Weigh: Interleaved Head Attention for Multimodal Physiological Stress Recognition in the Wild." *Published at **ECCV 2026 Workshop** on Affective & Behavior Analysis in-the-Wild (ABAW).* **[Paper]**

## Research Experience

**Research Intern, Computational Pathology Laboratory** *Jul 2026 – Present*
*Advanced Centre for Treatment, Research and Education in Cancer (ACTREC), Tata Memorial Centre, India*
*Research Outcome: Accepted at the NeurIPS 2026 Cancer Pathology Foundation Models Workshop.* [Paper]

- Developed a weakly supervised multiple-instance learning framework for predicting rare joint molecular phenotypes from H&E whole-slide images using pathology foundation models.
- Designed biomarker-specific representation learning with explicit interaction modeling and structured molecular-state supervision to distinguish joint-positive, single-positive, and double-negative phenotypes.
- Conducted comparative evaluations and component ablations across colorectal and lung adenocarcinoma cohorts using UNI2-h and CONCH, including analysis of molecular-state discrimination and spatial attention patterns.

**Project Research Assistant, Medical College of Wisconsin, USA (Remote)** *Feb 2025 – Present*

- Developed an EEG de-identification tool to support privacy-preserving preparation and organization of heterogeneous clinical recordings for multi-site research.
- Designing EEG preprocessing strategies to accommodate variations in channel counts, electrode configurations, and acquisition protocols across clinical sites, reducing reliance on fixed electrode montages.
- Investigating the use of EEG foundation models and clinically informed signal patterns for focal and generalized seizure detection, with a focus on reducing model complexity, improving cross-site generalization, and enabling efficient deployment in clinical settings for patients aged 10 years and older.

**Research Associate, E-Healthcare Research Centre (EHRC), IIIT Bangalore** *Aug 2024 – Dec 2025*

- Extended DeepNucleiNet research through sensitivity analyses across varying proportions of necrotic tissue within image tiles, characterizing detection limitations in partially necrotic regions.
- Conducted region-wise and areal analyses across 15 pathologist-annotated necrotic regions, examining the effects of region size, tissue coverage, and interior versus boundary tiles on necrosis localization.
- Investigated the sensitivity of necrosis detection to WSI tile-grid positioning using five spatially shifted extraction grids, combining predicted regions to reduce dependence on grid alignment and mitigate missed detections.

**Research Project Collaborator, Neuropathology Lab** *Sept 2021 – Aug 2023*
*National Institute of Mental Health and Neurosciences (NIMHANS), Bengaluru*

- Processed and curated gigapixel H&E whole-slide images of meningioma, glioblastoma, and low-grade glioma for computational pathology research.
- Collaborated with neuropathologists at the NIMHANS Neuropathology Laboratory on histological annotations, biologically informed feature design, and validation of necrosis detection methods.
- Developed software tools to streamline WSI annotation, automate tile-level label generation, and support pathology dataset preparation.

**Research Associate, Mirai Research Innovation Institute, Japan** *Oct 2020 – Aug 2021*

- Developed a CNN-based visual inspection pipeline using ResNet50 and transfer learning for detecting missing electronic components in PCB imagery, improving classification accuracy from **82% to 94%**.

## Technical Skills

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| **Digital Pathology** | Gigapixel WSI processing, OpenSlide, QuPath, ImageScope, StarDist |
| **Tissue & Spatial Analysis** | Nuclear segmentation and morphology, spatial tissue organisation, cell phenotyping, tumour microenvironment analysis |
| **Foundation Models & MIL** | UNI2-h, Virchow2, CONCH, weakly supervised learning, multiple-instance learning |
| **Deep Learning & Vision** | PyTorch, TensorFlow, Vision Transformers, visual representation learning, domain adaptation |